

Are the ground reaction forces altered by the curve and with the increasing sprinting velocity?

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Abstract

In 200- and 400-m races, 58% of the total distance to cover is in the curve. In the curve, the sprinting performance is decreased in comparison to the straight. However, the reasons for this decreased performance is not well understood. Thus, the aim of this study was to identify the kinetic parameters underpinning the sprinting performance in the curve in comparison to the straight. Nineteen experienced-to-elite curve specialists performed five sprints in the straight and in the curve (radius 41.58 m): 10, 15, 20, 30, and 40 m. The left and the right vertical, anterior–posterior, medial–lateral, and resultant ground reaction forces (respectively F_V , F_{A-P} , F_{M-L} , and F_{TOT}), the associated impulses (respectively IMP_V , IMP_{A-P} , IMP_{M-L} , and IMP_{TOT}) and the stance times of each side were averaged over each distance. In the curve, the time to cover the 40-m sprint was longer than in the straight (5.52 ± 0.25 vs. 5.47 ± 0.23 s, respectively). Additionally, the left and the right F_{A-P} and IMP_{A-P} were lower than in the straight while the left and the right F_{M-L} increased, meaning that the F_{M-L} was more medial. The left F_V was also lower than in the straight while the left stance times increased to keep the left IMP_V similar to the straight to maintain the subsequent swing time. Overall, the sprinting performance was reduced in the curve due to a reduction in the left and the right F_{A-P} and IMP_{A-P} , that were likely attributed to the concomitant increased F_{M-L} to adopt a curvilinear motion.

KEYWORDS

curve sprinting, force platforms, kinetics, performance, sprint

1 | INTRODUCTION

In straight-line sprinting, several studies reported the necessity to apply a large magnitude of mean anterior–posterior force (F_{A-P}) and anterior–posterior impulse (IMP_{A-P}) to improve the sprinting performance.^{1–6} Nevertheless, for all track events starting from the 200 m, the curve portion of the track represents almost 58% of the total distance to cover,⁷ which makes the curve inextricably linked to performance for these

events. In this sprinting condition, the sprinting performance decreases due to the reduced running velocities: -2.3% at 13 m^8 and -4.7% at 40 m^9 in comparison to the straight.

Hence, to understand the underlying reasons for the decreased performance in the curve, investigating the ground reaction forces (GRF) in this sprinting condition seems of great importance.

Thus far, some mathematical models have tried to address the causes of the reduced performance in the

curve.^{10–12} Greene,¹¹ followed by Usherwood and Wilson,¹² speculated that humans sprinting in the curve would produce the same peak resultant force (F_{TOT}), yet with a different distribution of the three GRF orthogonal components which was referred as the “constant limb hypothesis”.¹² Briefly, they postulated that the vertical force (F_V) would decrease, concomitantly with an increase in the medial-lateral force (F_{M-L}), necessary to produce the centripetal force and maintain the curved trajectory.¹¹ These models further hypothesized that the stance times would in turn increase to maintain an IMP_V similar to the straight.^{11,12}

Experimentally, only two studies have investigated the GRF^{8,13} in the curve at radii typical of track events and with maximal-effort sprints. Nevertheless, their analyses were limited at two specific instants of the sprint: at 13⁸ and 40 m.¹³ Using a Statistical Parametric Mapping method, Judson et al.⁸ showed that the left (i.e., inside) and the right (i.e., outside) F_{M-L} were more medial during the majority of the stance in the curve in comparison to the straight 13 m after the starting blocks exit. However, they did not investigate neither the F_V , nor the F_{A-P} . At 40 m, Churchill et al.¹³ reported reduced left peak F_{TOT} and F_V in the curve, due to the large increase in the left peak F_{M-L} . Concomitantly, the left stance times increased in comparison to the straight to produce a similar left IMP_V .¹³ In contrast, no modifications in the right F_{TOT} and F_V were reported,¹³ suggesting that only the left (i.e., inside) limb is affected by the curve. Consequently, this study showed that the “constant limb hypothesis” failed for the left limb only in the curve. However, the authors did not report the F_{A-P} . Besides, as the centripetal force is strongly influenced by the sprinting velocity,¹¹ the GRF in the curve should be compared to the straight throughout the entire acceleration phase (i.e., from the starting blocks to the maximal anterior–posterior velocity ($V_{A-P MAX}$)) to comprehend whether the increasing anterior–posterior velocity (V_{A-P}) impacts the GRF in the curve. Hence, a clear gap remains in the literature regarding the curve and the GRF in this sprinting condition must be analyzed more thoroughly.

Thus, the main aim of this study was to identify the kinetic parameters characterizing the decreased sprinting performance in the curve from the starting blocks pushing phase to the $V_{A-P MAX}$.

It was hypothesized that the mean F_{M-L} would increase in the curve, with a concomitant decrease in the mean F_V and F_{A-P} , and the F_{TOT} in the curve would remain similar to the straight. A secondary hypothesis of the present study was that, because of the reduced mean F_V in the curve, the stance times would increase to produce a IMP_V similar to the straight.

2 | METHODS

2.1 | Participants

Nineteen (15 male and four female) experienced-to-elite curve sprinters (mean $\pm$ SD: age = 23.9 ± 3.7 years; body mass = 73.9 ± 7.4 kg; height = 1.78 ± 0.07 m) volunteered to participate in this study. The six male 200-m specialists mean personal best (PB) was 22.89 s (range 20.45–23.83 s), the nine male 400-m specialists mean PB was 49.56 s (range 47.60–53.31 s), the two female 200-m specialists mean PB was 26.3 s (range 24.85–27.75 s) and the two female 400-m specialists mean PB was 57.67 s (57.40–57.93 s). After an explanation of the protocol, the athletes gave their written informed consent to participate to the experimentation, conducted in accordance with the declaration of Helsinki and approved by the local ethical committee (IRB00012476-2021-29-04-107).

2.2 | Experimental setup

This experimentation took place in an indoor track facility. The instantaneous F_V , F_{A-P} , and F_{M-L} were measured by a 6.60-m long force platform (FP) (see Figure 1), composed out of six individual FP (five length-wise and one sideways; 1.2×0.6 m each sampling at 1000 Hz; KI 9067; Kistler, Wintherthu, Switzerland) connected in-series and embedded within the track surface. The second FP was oriented perpendicularly so that the participants' hands were positioned on the FP during the starting blocks phase (see Figure 1).

For the curve sprints, we reconstructed a lane five (radius = 41.58 m) overlapping the FP area by positioning two white bands (DimaSport, Ozoir-la-Ferrière, France; see the dark purple lines in the Figure 1) on the track to replicate the inner and the outer lanes of a 1.22-m wide lane five.

2.3 | Protocol

The participants performed 10-, 15-, 20-, 30-, and 40-m sprints, with maximal effort, in the straight and in the curve with a randomized sprinting condition (straight and curve) order, to limit the fatigue effect. They followed the “on your marks”, “set”, and “go” instructions using starting blocks and were wearing their own spikes. The participants were given 3, 4, 5, 6, and 8 min of rest after the 10-, 15-, 20-, 30-, and 40-m sprints, respectively, in order to avoid the onset of fatigue.

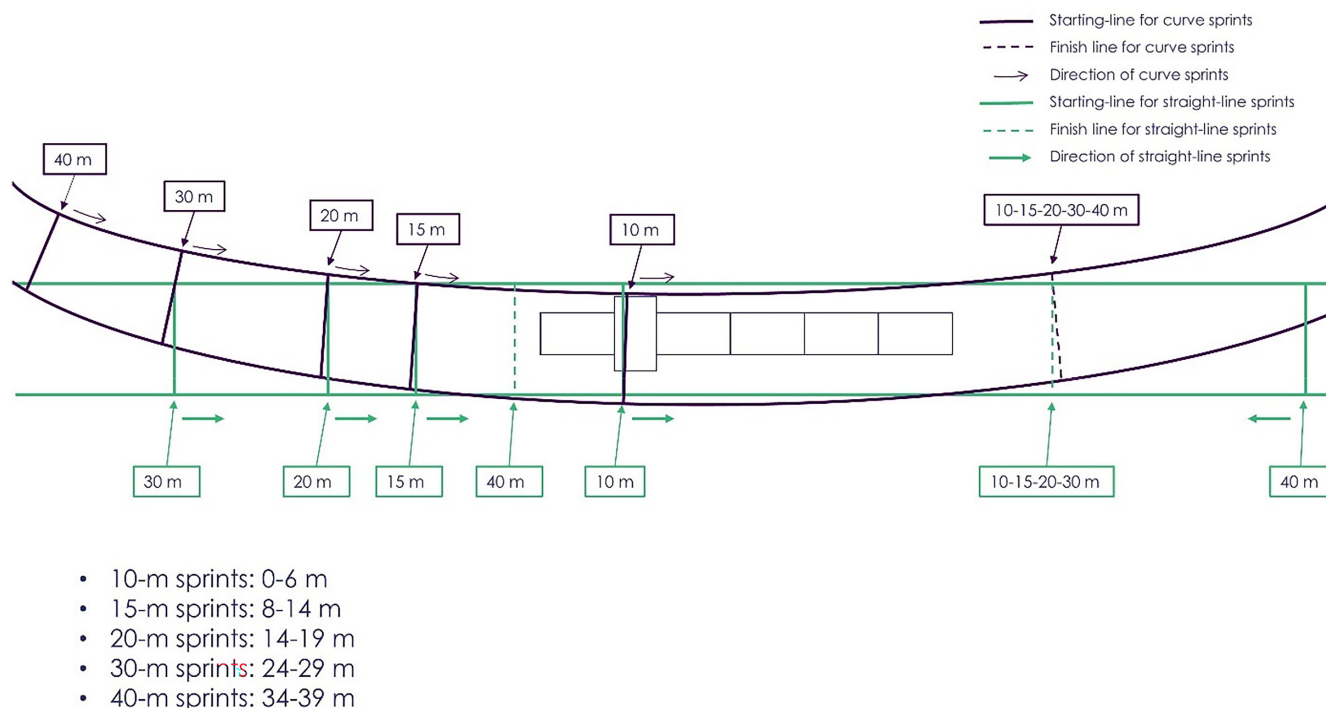


FIGURE 1 Graphical representation of the protocol used to collect the ground reaction forces (GRF) at different instants of the race. Overall, with the increasing distance, the starting blocks position was positioned further away from the force platforms in order to collect the GRF of the subsequent stances. Note that for the 40-m sprints, the starting blocks were positioned on the other side and the participants were sprinting on the opposite way due to the indoor facility organization. The dimensions are not at a correct scale.

To compute the GRF over multiple stances, we adapted a protocol based on previous research^{6,14} (see the [Figures 1](#) and [2](#)). For the 10-m sprints, the starting blocks were positioned on the first FP (see [Figure 1](#)). For the subsequent sprints, the starting blocks were progressively positioned further away from the FP so that the GRF of consecutive stances could be computed (see [Figure 1](#)). This methodology permitted to collect the GRF of ~17 stances in average in the straight and ~15 stances in average in the curve.

2.4 | Data processing

The entire data processing detailed in the following paragraphs was realized using a customized Matlab (R2017b MathWorks Inc. Natick, MA, USA) script. The raw GRF data were low-pass filtered, third-order, zero-phase shifting (forward-backward filtering), Butterworth filter with a cut-off frequency of 200 Hz, chosen after residual analysis.¹⁵

2.4.1 | Sprinting performance

Two pairs of timing gates (Witty, Microgate, Bolzano, Italy) were positioned at the beginning and at the end of each sprinting distance. The first pair of timing gates was placed

~10 cm ahead of the starting line such that the participants were not obstructing the beam while on the starting blocks. The sprinting performance (expressed in s) over the different sprints corresponded to the elapsed time over each distance measured with the two pairs of timing gates.

2.4.2 | Video cameras

To make sure that the participants' feet made contact strictly within the FP, each sprint was filmed using two video cameras (120 Hz, iPhone 11, Apple, California, USA) positioned on either side of the FP. A stance was considered "valid" when the foot fully landed within the FP area and was used for further analyses.

2.4.3 | Events detection

The beginning and the end of the starting blocks pushing phase were defined as the instant when the change in the F_V rose above 15 N of the body weight (BW) and when the F_V first dropped below 20 N, respectively.⁶ After the starting blocks exit, the sprint events (touchdown and toe-off) were determined using a 20-N threshold on the F_V .^{16,17}

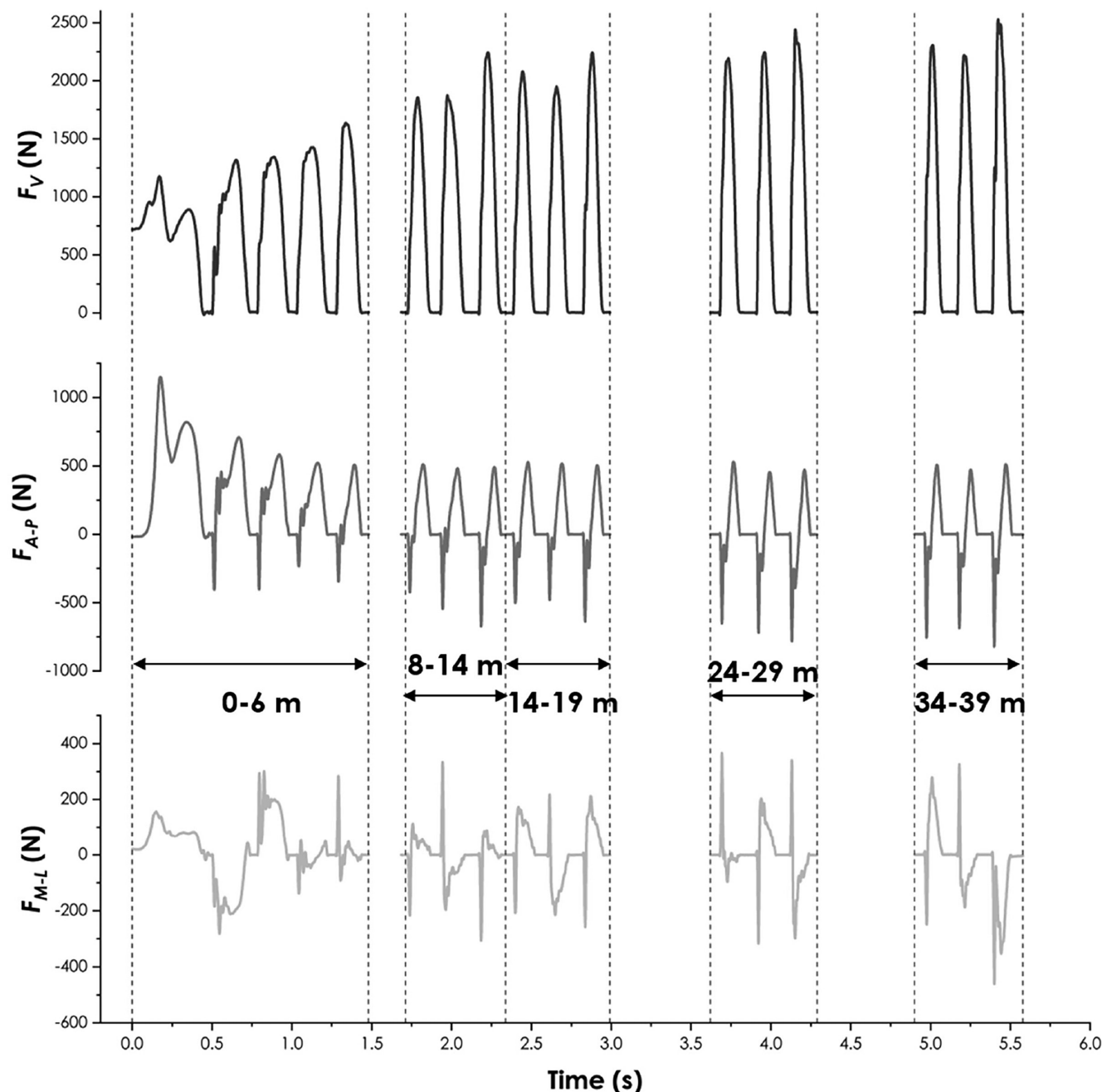


FIGURE 2 Typical signals of the instantaneous vertical (F_V), anterior-posterior (F_{A-P}) and medial-lateral (F_{M-L}) components of the ground reaction force (GRF) obtained during the 10-, 15-, 20-, 30-, and 40-m sprints in the straight that correspond to the following location of the force platforms area: (A) 10-m sprints: Zone 1 (0–6 m; 0 corresponding to the starting line); (B) 15-m sprints: Zone 2 (8–14 m); (C) 20-m sprints: Zone 3 (14–19 m); (D) 30-m sprints: Zone 4 (24–29 m) and 40-m sprints: Zone 5 (34–39 m).

2.4.4 | Stance times

The starting blocks pushing times (expressed in ms) were computed as the elapsed time between the beginning and the end of the starting blocks pushing phase. Afterwards, the left and the right stance times (expressed in ms) were computed as the elapsed time from the ipsilateral touchdown to the following ipsilateral take-off. Finally, we averaged the left and the

right stance times separately over each zone (five zones that correspond to 0–6, 8–14, 14–19, 24–29, and 34–39 m; see [Figures 1 and 2](#)).

2.4.5 | Global coordinate system rotation

In the straight, the experimentations generally assumed that the anterior-posterior component of the FP global

coordinate system (GCS) was aligned with the straight-line lane.^{1,2,4,6,16,18–21,22} In our study, the anterior–posterior component of the FP was also aligned with the straight-line lane (see Data S1 for the entire GCS rotation processing details). In the curve, the anterior–posterior component of the FP was aligned with the curve lane by calculating the angle (θ) between a vector following the FP antero-posterior axis (the black arrow) and the reconstructed curve²³ at each 0.10 m along the curve. Afterwards, at each stance, the FP GCS was aligned with the curve lane by rotating the F_{A-P} and the F_{M-L} about the vertical axis.

$$\begin{bmatrix} F'_{A-P} \\ F'_{M-L} \end{bmatrix} = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} F_{A-P} \\ F_{M-L} \end{bmatrix} \quad (1)$$

Where F_{A-P} and F_{M-L} are GRF outputs from the FP, F'_{A-P} and F'_{M-L} correspond respectively to the rotated F_{A-P} and F_{M-L} at each stance. For clarity's sake, F'_{A-P} and F'_{M-L} will be referred to F_{A-P} and F_{M-L} respectively for the rest of this manuscript.

2.4.6 | GRF forces and impulses

The left and the right F_{TOT} , F_V , F_{A-P} , and F_{M-L} were averaged over each “valid stance” (i.e., with the foot fully within the FP) and expressed relative to the body mass (thus expressed in $N\ kg^{-1}$). The left and the right IMP_{TOT} , IMP_V , IMP_{A-P} , and IMP_{M-L} were computed as the product of the stance time and the F_{TOT} , F_V , F_{A-P} , and F_{M-L} , respectively. The IMP_V was computed by subtracting the impulse due to the BW. The impulses were normalized to the body mass (thus expressed in ms^{-1}), in order to reflect the changes in the CM velocity.^{1,2,4,8,13} For both F_{M-L} and IMP_{M-L} , a positive value indicates a medial F_{M-L} and IMP_{M-L} while a negative value indicates a lateral F_{M-L} and IMP_{M-L} .

2.4.7 | Data analysis

Thereafter, the F_{TOT} , F_V , F_{A-P} , and F_{M-L} , the IMP_{TOT} , IMP_V , IMP_{A-P} , and IMP_{M-L} were averaged over each zone. As an example, a participant performing a 10-m sprint had four valid stances after the starting blocks exit with the left foot in contact with the ground during the stances 1 and 3; and the right foot in contact with the ground for the stances 2 and 4. Thus, the mean left F_{TOT} , F_V , F_{A-P} , and F_{M-L} at 0–6 m corresponded to the mean F_{TOT} , F_V , F_{A-P} , and F_{M-L} over the 2 left stances. On the other hand, the mean right F_{TOT} , F_V , F_{A-P} , and F_{M-L} at 0–6 m corresponded to the mean F_{TOT} , F_V , F_{A-P} , and F_{M-L} over the 2 right stances.

The mean left and right F_{TOT} , F_V , F_{A-P} , and F_{M-L} , stance times, IMP_{TOT} , IMP_V , IMP_{A-P} , and IMP_{M-L} over the starting blocks pushing phase and through the five next zones that corresponded to 0–6, 8–14, 14–19, 24–29, and 34–39 m were used for further analysis and compared between the straight and the curve.

2.4.8 | Statistical analyses

The statistical analyses were performed using JASP (0.15.0.0). The normal distribution of the data was checked for each variable using the Shapiro–Wilk normality test. During the starting blocks pushing phase, the mean left and right starting blocks pushing phase duration and the mean F_{TOT} , F_V , F_{A-P} , F_{M-L} , IMP_{TOT} , IMP_V , IMP_{A-P} , and IMP_{M-L} were compared between the straight and the curve using paired t -tests. During the acceleration phase, the effects of the condition (straight vs. curve), the effects of the distance (0–6, 8–14, 14–19, 24–29, and 34–39 m) and the condition $\times$ distance interactions of the mean F_{TOT} , F_V , F_{A-P} , F_{M-L} , stance times, IMP_{TOT} , IMP_V , IMP_{A-P} and IMP_{M-L} were tested using two-way analyses of variance (ANOVA) with repeated measures. The homogeneity and the sphericity (Mauchly's test) were also checked and a Geisser–Greenhouse correction applied when necessary. The effect sizes were described with the omega squared (ω^2 , with $\omega^2 < 0.01$; $\omega^2 > 0.01$; $\omega^2 > 0.06$, and $\omega^2 > 0.14$ indicating respectively trivial, small, medium and large effect sizes). For all the statistical analyses, the alpha level was set as $\alpha = 0.05$. The values are reported as mean $\pm$ SD unless otherwise stated.

3 | RESULTS

3.1 | Sprinting performance

The sprinting performance during the 10-m sprints was similar in the straight and in the curve (see Table 1). However, starting from the 15-m sprint onwards, the sprinting performance in the curve was significantly poorer than in the straight (all $p < 0.026$, see Table 1).

3.2 | Ground reaction forces

3.2.1 | The starting blocks pushing phase

During the starting blocks pushing phase in the curve, the mean F_{M-L} was more lateral than in the straight (respectively -0.26 ± 0.57 vs. $-0.06 \pm 0.53\ N\ kg^{-1}$; $p = 0.002$). In contrast, no differences were observed for the mean F_{TOT} , F_V , and F_{A-P} (all $p \geq 0.712$) (see Figure 3).

TABLE 1 Mean ($\pm$ SD) sprinting performance (in s) in the straight and in the curve from the different sprinting distances measured with photocells.

Sprints (m)	Sprinting performance in the straight (s)	Sprinting performance in the curve (s)
10m	1.98 $\pm$ 0.06	1.99 $\pm$ 0.07
15m	2.63 $\pm$ 0.09	2.65 $\pm$ 0.10*
20m	3.22 $\pm$ 0.12	3.25 $\pm$ 0.14*
30m	4.37 $\pm$ 0.19	4.40 $\pm$ 0.21*
40m	5.47 $\pm$ 0.23	5.52 $\pm$ 0.25*

*Indicates significantly different from the straight ($p < 0.05$).

3.2.2 | Acceleration phase

During the acceleration phase, the left F_{TOT} ($F_{(1,12)} = 5.717$, $p = 0.034$, $\omega^2 = 0.024$) and the left F_V ($F_{(1,12)} = 8.018$, $p = 0.015$, $\omega^2 = 0.036$) were lower in the curve than in the straight (see Figure 4).

Both the left ($F_{(1,12)} = 45.001$, $p < 0.001$, $\omega^2 = 0.214$) and the right F_{A-P} ($F_{(1,8)} = 24.973$, $p = 0.001$, $\omega^2 = 0.143$) were lower in the curve in comparison to the straight (see Figure 4). In addition to that, we observed a significant condition $\times$ distance interaction for the left ($F_{(1,12)} = 7.394$, $p < 0.001$, $\omega^2 = 0.145$) and the right F_{A-P} ($F_{(1,8)} = 2.861$, $p = 0.039$, $\omega^2 = 0.055$). The post-hoc analyses with the Bonferroni corrections showed

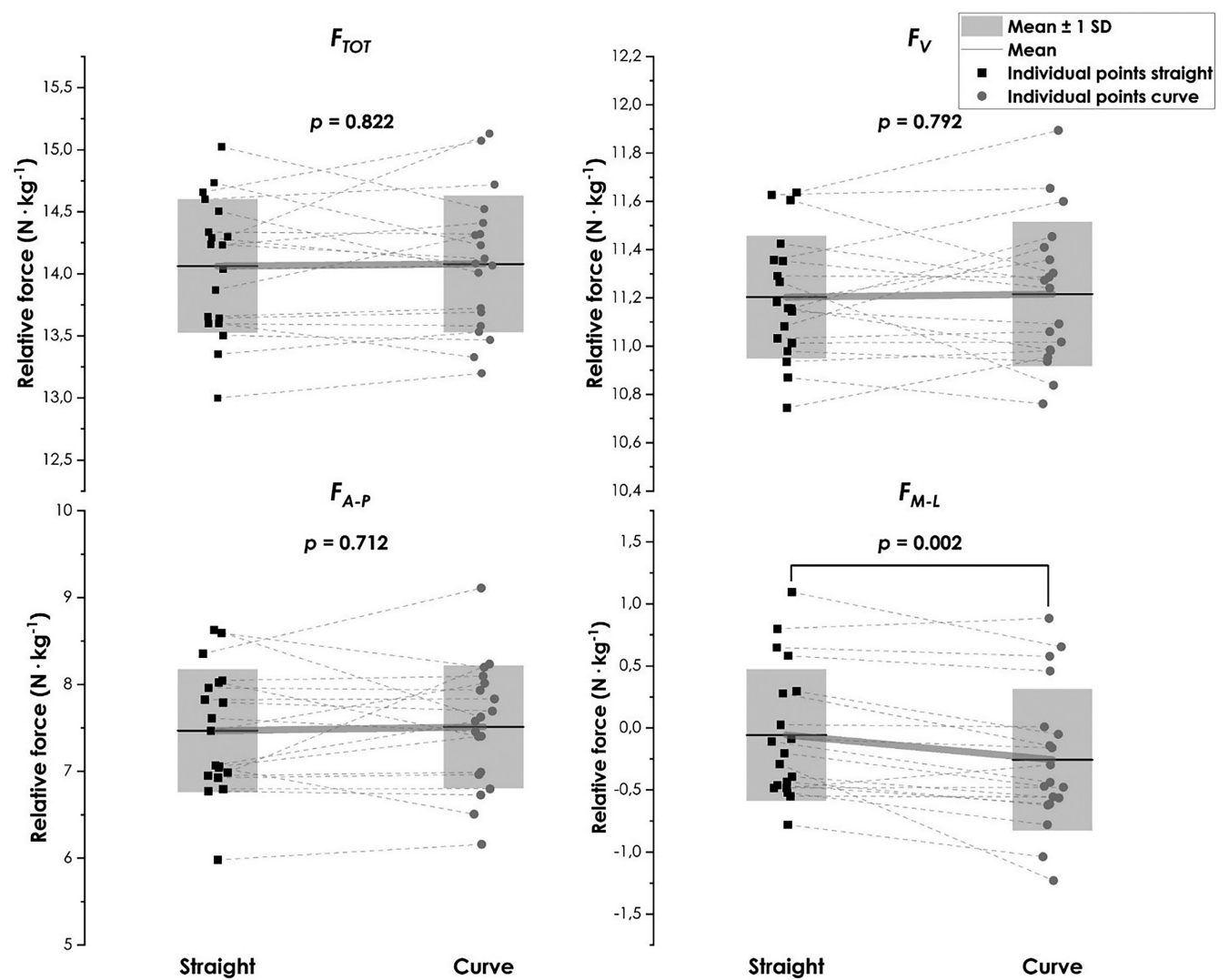


FIGURE 3 Individual values (each dot represent the mean value of a participant) of the resultant (F_{TOT}), vertical (F_V), anterior-posterior (F_{A-P}), and medial-lateral (F_{M-L}) relative to body mass (expressed in $N \cdot kg^{-1}$) averaged over the starting blocks pushing phase. The bold traces connect the mean values of the whole sample ($n = 19$) in the two sprinting conditions. The box charts correspond to mean $\pm$ 1 SD and the dashed traces connect individual values between the two sprinting conditions.

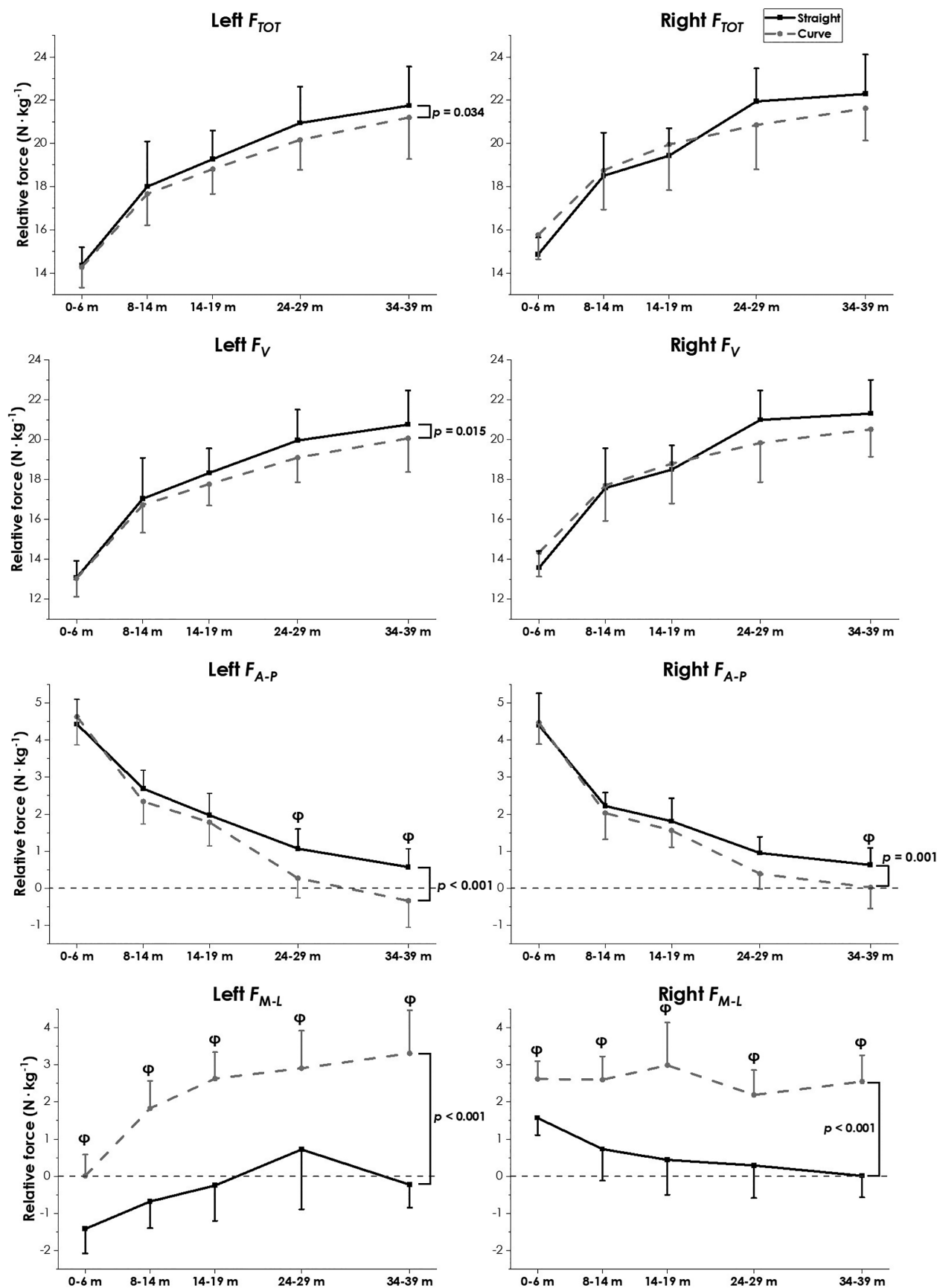


FIGURE 4 Mean ($\pm$ SD) left and right resultant (F_{TOT}), vertical (F_V), anterior-posterior (F_{A-P}) and medial-lateral (F_{M-L}) ground reaction forces (GRF) averaged during the stances in the straight (black) and in the curve (dashed gray) over each zone. For all panels, the brackets indicate a significant main effect for condition ($p < 0.05$). Finally, ϕ indicates a significant condition $\times$ distance interaction ($p < 0.05$).

that the left F_{A-P} was lower in the curve than in the straight for the left stances starting from 24 to 29 m onwards. On the other hand, the right F_{A-P} was lower in the curve than in the straight at 34–39 m (see Figure 4).

The left ($F_{(1,10)}=487.54$, $p<0.001$, $\omega^2=0.846$) and the right F_{M-L} ($F_{(1,8)}=438.822$, $p<0.001$, $\omega^2=0.779$) were more medial in the curve than in the straight (see Figure 4). Additionally, a significant condition $\times$ distance interaction were found for the left ($F_{(1,10)}=8.716$, $p<0.001$, $\omega^2=0.202$) and the right F_{M-L} ($F_{(1,8)}=5.727$, $p=0.009$, $\omega^2=0.143$) at each distance. Consequently, within the curve, the left and the right F_{M-L} were more medial than in the straight at each zone (all $p\leq 0.003$; see Figure 4).

3.3 | Stance times

3.3.1 | Starting blocks pushing phase

No differences were observed between the straight and the curve for the starting blocks pushing time (401 ± 25 vs. 404 ± 29 ms, respectively; $p=0.556$).

3.3.2 | Acceleration phase

After the starting blocks exit, the left stance times were longer in the curve than in the straight ($F_{(1,11)}=42.009$, $p<0.001$, $\omega^2=0.112$; see Figure 5). In contrast, no differences were observed for the right stance times between the straight and the curve ($F_{(1,12)}=0.224$, $p=0.646$, $\omega^2=0.000$; see Figure 5).

3.4 | Impulses

3.4.1 | Starting blocks pushing phase

During the starting blocks pushing-phase, the relative IMP_{M-L} was more lateral in the curve than in the straight (-0.11 ± 0.23 vs. -0.02 ± 0.21 ms⁻¹; $p=0.001$; see Figure 6). No other significant differences were observed (all $p\geq 0.140$; see Figure 6).

3.4.2 | Acceleration phase

The left ($F_{(1,12)}=39.808$, $p<0.001$, $\omega^2=0.221$) and the right IMP_{A-P} ($F_{(1,8)}=42.257$, $p<0.001$, $\omega^2=0.165$) were lower in the curve than in the straight (see Figure 7). Additionally, there was a significant condition $\times$ distance interaction for the left IMP_{A-P} ($F_{(1,12)}=6.945$, $p<0.001$, $\omega^2=0.145$) indicating that the left IMP_{A-P} was lower in the curve than in the straight starting from 24 to 29 m onwards ($p<0.001$; see Figure 7).

The left ($F_{(1,12)}=453.279$, $p<0.001$, $\omega^2=0.861$) and the right IMP_{M-L} ($F_{(1,8)}=341.636$, $p<0.001$, $\omega^2=0.792$) were more medial in the curve than in the straight (see Figure 7). Lastly, we observed significant condition $\times$ distance interactions for the left ($F_{(1,12)}=5.080$, $p<0.001$, $\omega^2=0.112$) and the right IMP_{M-L} ($F_{(1,8)}=3.975$, $p=0.033$, $\omega^2=0.101$). The post-hoc analyses with the Bonferroni corrections showed that, at each distance, the left and the right IMP_{M-L} in the curve were more medial than the left and right IMP_{M-L} in the straight ($p\leq 0.001$; see Figure 7).

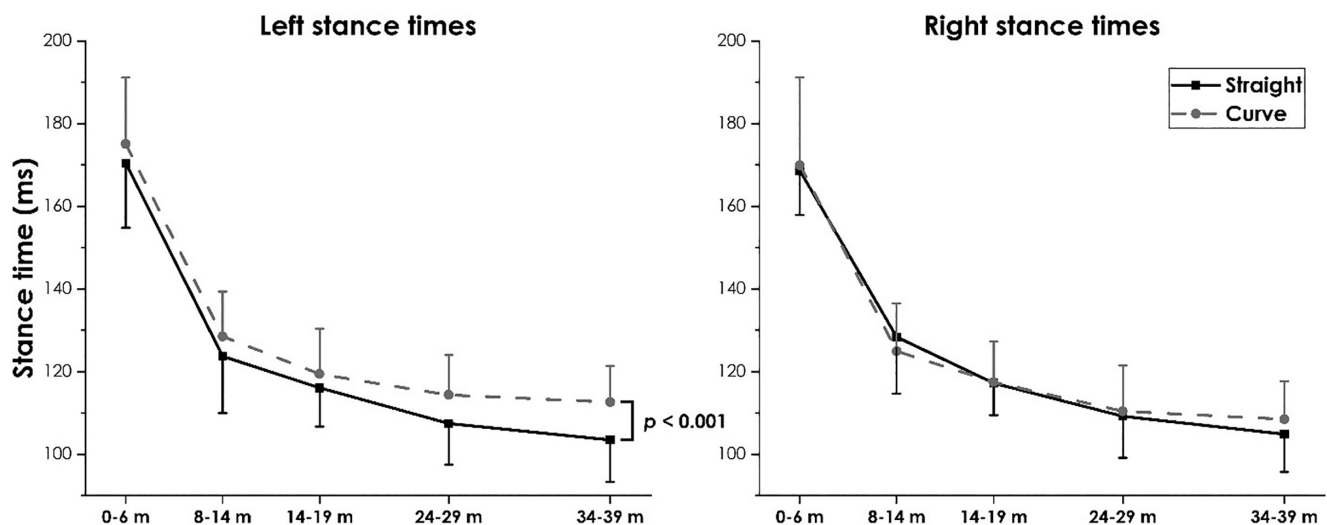


FIGURE 5 Mean ($\pm$ SD) left and right stance times (expressed in ms) over each interval in the straight (black) and in the curve (dashed gray). The brackets indicate a significant main effect for condition ($p<0.05$).

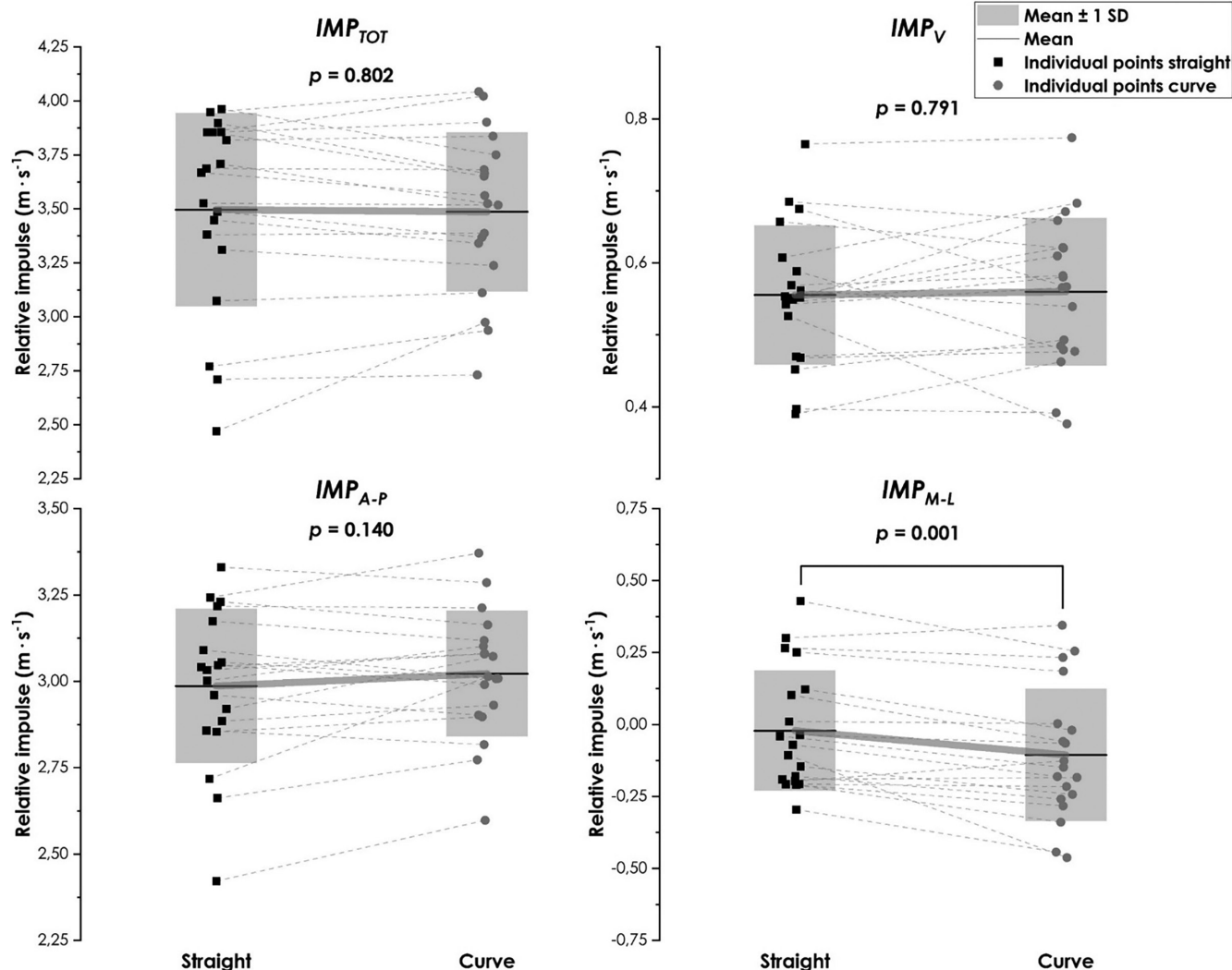


FIGURE 6 Individual values (each dot represent the value of a participant) of the resultant (IMP_{TOT}), vertical (IMP_V), anterior-posterior (IMP_{A-P}), and medial-lateral (IMP_{M-L}) impulses expressed relative to body mass (in $m \cdot s^{-1}$) over the starting blocks pushing phase. The bold traces connect the mean of the whole sample ($n = 19$) between the straight and the curve. The box charts correspond to the mean ± 1 SD and the dashed traces connect the individual values between the straight and the curve.

4 | DISCUSSION

The core novelty of the present experimentation was the ability to explore the GRF over 15–19 foot contacts in the curve (from the starting blocks to the $V_{A-P \text{ MAX}}$) while previous studies investigated the GRF in the curve during a single stance at $13 m^8$ or at $V_{A-P \text{ MAX}}$.¹³ In the present study, we found that the sprinting performance was reduced (see Table 1) in the curve due to the decreased left and right mean F_{A-P} in the curve, likely due to the increased left and right F_{M-L} and as the participants were unable to increase their F_{TOT} .

The mean F_{M-L} was more lateral in the curve than in the straight during the starting blocks pushing phase (respectively -0.26 ± 0.57 vs. $-0.06 \pm 0.53 N \cdot kg^{-1}$; see Figure 3), meaning that, at the group level, the participants were heading towards the outside of their lane (i.e., laterally) after the starting blocks exit. In the present experimentation, 14 out

of 19 participants started with their left foot on the front block. Among them, 13 participants produced a lateral mean F_{M-L} and IMP_{M-L} over the starting blocks pushing phase in the curve, meaning that the participants were heading towards the outside of their lane. In contrast, four out of the five participants that started with their right leg on the front block produced a medial F_{M-L} and IMP_{M-L} in the curve, meaning that the participants were heading towards the inside of their lane. These findings further question the choice of the front and rear foot in the starting blocks in the curve as this choice will in turn impact the sprinter's direction after the starting blocks exit, together with the starting-blocks position in the lane.

Indeed, in the present experimentation, the starting blocks were positioned on the middle of the lane as the FP were narrower than the lane (0.60 vs. 1.22 m, respectively) which represents a limitation of the present study and might have impacted the findings. Hence,

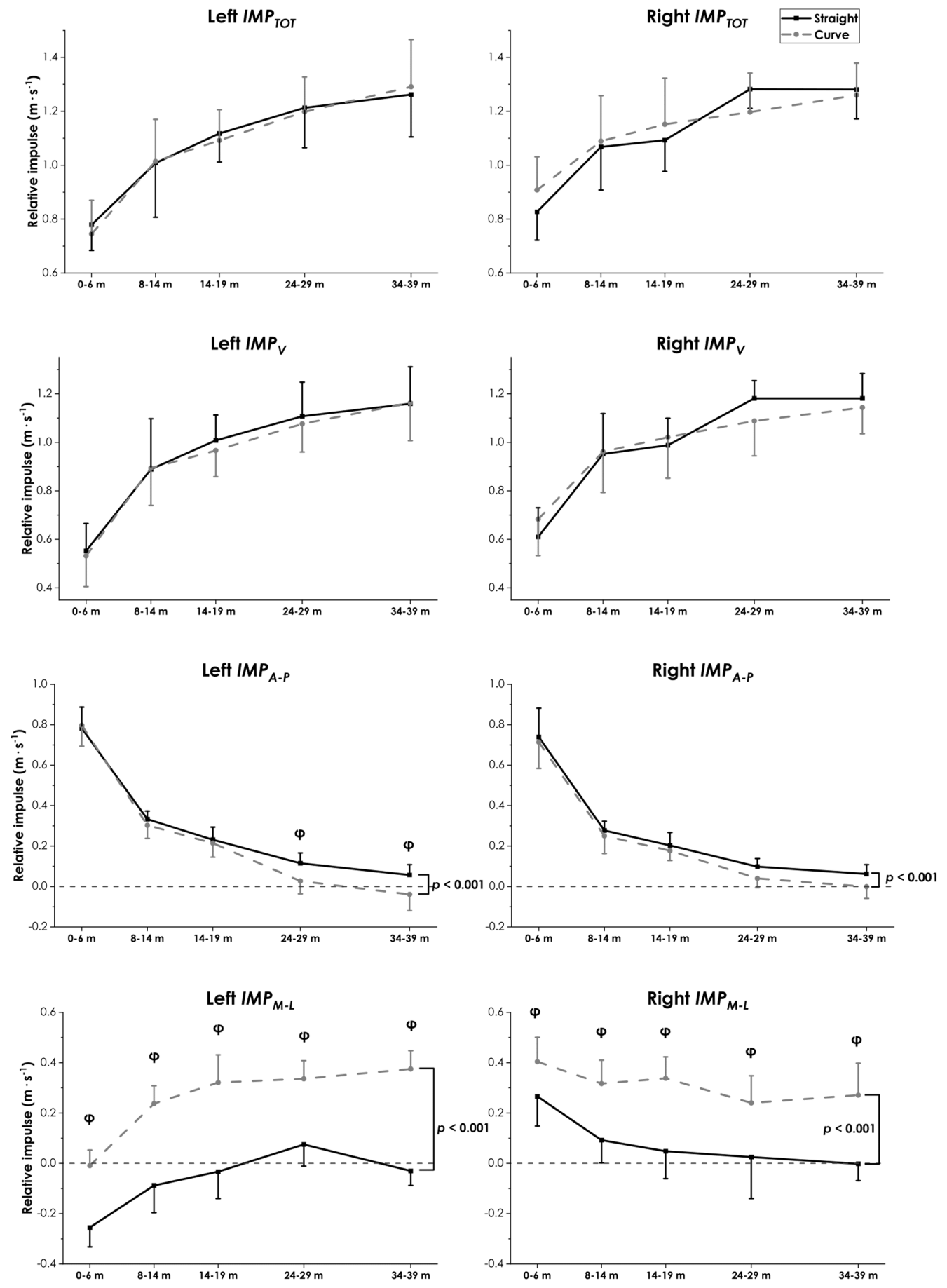


FIGURE 7 Mean ($\pm$ SD) resultant (IMP_{TOT}), vertical (IMP_V), anterior–posterior (IMP_{A-P}) and medial-lateral (IMP_{M-L}) impulses for the left and right stances in the straight (black) and in the curve (dashed gray) averaged over each distance. For all panels, the brackets indicates a significant main effect for condition ($p < 0.05$). Finally, ϕ indicates a significant condition $\times$ distance interaction ($p < 0.05$).

we can also hypothesize that the participants with the left foot on the front block produced lateral F_{M-L} and IMP_{M-L} in the curve to move away from the inner line, before heading medially during the stances that follow, which is eventually unlikely to be optimal for the sprinting performance. Rather, in curve sprinting events, athletes usually position their starting blocks on the outside of the lane, pointing out towards a point, tangent to the inner part of the lane, so the athletes do not need to move away from the inner line. Using this set-up, the starting blocks pushing phase and the initial steps that follow are roughly in straight-line which is thought to be more optimal for the sprinting performance. Hence, it would be interesting to investigate the GRF in the curve with the starting blocks positioned on the outside of the lane in a future experimentation.

After the starting blocks exit, we reported a large increase in the left and the right F_{M-L} in comparison to the straight. The mean F_{M-L} was positive, meaning that the participants were heading medially, contrary to the starting blocks pushing phase, and to adopt a curvilinear motion. Concomitantly, we observed a decreased F_{A-P} as the participants were unable to increase the F_{TOT} . As the F_{A-P} is key for better sprinting performance⁶ we can assume that, in the curve, the sprinting performance is, at least partially, decreased due to the reduced left and right F_{A-P} .

Besides, we showed that the left mean F_{TOT} was lower in the curve than in the straight (see Figure 4). Therefore, in line with Churchill et al.'s¹³ findings, we can further reject the “constant limb hypothesis” that assumes that the F_{TOT} would remain similar in the curve in comparison to the straight,¹² for the left limb, and throughout the acceleration phase. Previously, Alt et al.²⁴ showed that the left ankle eversion increased in the curve in comparison to the straight. Additionally, Judson et al.⁸ showed that, in the curve, the left foot push-off axis switched from transverse to oblique. Hence, we can hypothesize that the ability to produce a left F_{TOT} in the curve similar to the straight (i.e., the constant limb hypothesis) could be altered by the alterations of the left ankle and foot kinematics in the curve.

In contrast, the right mean F_{TOT} and F_V were not altered in the curve (see Figure 4), which is consistent with what was previously reported in lane 2¹³ or at tight radii.²⁵ For the right limb, only the horizontal plane was affected whereby we observed an increase in the mean F_{M-L} concomitantly with a decrease in the mean F_{A-P} in comparison to the straight (see Figure 4).

No changes were observed between the straight and the curve for the starting blocks pushing phase duration.

Hence, as the F_{A-P} were similar between the straight and the curve, the IMP_{A-P} were also similar in the straight and in the curve with an anteroposterior velocity at the starting blocks exit of $\sim 3 \text{ ms}^{-1}$. These findings are consistent with the literature with groups of similar expertise²⁶ and slightly below the values of $3.48 \pm 0.05 \text{ ms}^{-1}$ reported among elite sprinters.²⁷ On the other hand, as the F_{M-L} were more lateral in the curve, the IMP_{M-L} were also more lateral in the curve during the starting blocks pushing phase.

After the starting blocks exit, the left mean stance times in the curve were longer than in the straight (see Figure 5). Consequently, our findings are consistent with what was previously reported at 13 m²⁸ and 40 m.^{9,13} On the other hand, no changes were reported for the right limb (see Figure 5), which is also similar to Churchill et al.'s.^{9,13} findings at 40 m. In the curve, the left limb stance times increased due to the lower left F_V and in an attempt to produce a similar IMP_V as in the straight,¹³ to maintain the following swing times and provide enough time for the opposite limb to be repositioned forward.

However, despite the increased left stance times in the curve, the left IMP_{A-P} remained lower than in the straight since the decreased left F_{A-P} was greater than the concomitant increased left stance times. The right IMP_{A-P} were also lower in the curve in comparison to the straight as the right F_{A-P} was lower in the curve and the stance times did not increase. Previously, the IMP_{A-P} was reported as a major parameter in overground sprinting performance.^{1,2,4} Thus, we can postulate that the sprinting performance in the curve is reduced due to the decreased left and right IMP_{A-P} and the increased left stance times that could in turn reduce the left step frequency.

In the present study, the mean F_{TOT} , IMP_{TOT} , F_V , and IMP_V increased progressively with the increasing distance both in the straight and in the curve, which is similar to what was previously reported in the existing literature in the straight.^{4,6,29} In contrast, the mean F_{A-P} and IMP_{A-P} decreased with the increasing distance both in the straight and in the curve, which is also consistent with what was previously reported in the straight.^{4,6,29} Consequently, the F_{TOT} , IMP_{TOT} , F_V , IMP_V , F_{A-P} , and IMP_{A-P} kinetics are similar in the straight and in the curve. Nonetheless, in the curve, the mean IMP_{A-P} became negative for both the left and the right limbs starting from 34 to 39 m while the mean IMP_{A-P} remained positive in the meantime in the straight (see Figure 7). Therefore, the participants reached their $V_{A-P \text{ MAX}}$ earlier in the curve, although this was previously reported as detrimental for the sprinting performance in the straight.^{30–32}

On the other hand, large discrepancies were observed regarding the mean F_{M-L} and the IMP_{M-L} between the straight and the curve. Indeed, in the straight, as Nagahara et al.³³ reported, the left and the right F_{M-L} and the IMP_{M-L} were opposite, though their sum equals zero so the participants' CM trajectory is eventually linear. In contrast, in the curve, the sum of the left and the right F_{M-L} and IMP_{M-L} were positive—thus medial—, which made the CM turn.

Additionally, the left and the right F_{M-L} and IMP_{M-L} behaved differently in the curve. Thus, although this was not a direct aim of the present study, we compared the left and the right F_{M-L} and IMP_{M-L} within the curve. The right mean F_{M-L} was $\sim 2.5 \text{ N kg}^{-1}$ at 0–6 m and plateaued afterwards (see Figure 4). On the other hand, the left mean F_{M-L} and IMP_{M-L} were close to zero at 0–6 m and subsequently increased progressively (see Figures 4 and 7). This means that during the first 14 m of the sprint in lane five, the right stances made the CM turn more than the left stances thanks to the participants' ability to generate greater F_{M-L} and IMP_{M-L} during the right stances which is consistent with Judson et al.'s⁸ findings stipulating that more turning of the CM occurred during the right stance at 13 m.

At 14–19 m, both the left and the right mean F_{M-L} and IMP_{M-L} were almost equal (see Figures 4 and 7), meaning that both limbs similarly contributed to the CM rotation. Starting from 24 to 29 m onwards, the left mean F_{M-L} and IMP_{M-L} became visually greater than the right F_{M-L} and IMP_{M-L} . Although we have not conducted a statistical analysis on this specific point, we can assume that, starting from 24 m onwards, the left stances apparently made the CM turn more than the right stances. Those findings are also consistent with what Churchill et al. reported at 40 m, stipulating that more turning of the CM occurred during the left stance at 40 m.⁹

5 | PERSPECTIVES

The present experimentation helped us to better comprehend the kinetic factors altering the sprinting performance in the curve. Overall, the left limb was the most affected in the curve and the sprinting performance was reduced in the curve mostly due to:

- The decreased left and right F_{A-P} and IMP_{A-P} that were likely caused by the concomitant increased left and right F_{M-L} and IMP_{M-L} , to adopt a curvilinear motion.
- The increased left stance times in the curve, likely due to the decreased F_{TOT} and F_V and in an attempt to produce a similar IMP_{TOT} and IMP_V in comparison to the straight, likely to maintain an adequate sprinting motion and balance.

The GRF and the impulses alterations in the curve eventually result from modifications in the sprinting technique (i.e., body kinematics) itself related to muscle activities. Therefore, investigating the lower-limbs kinematics and the muscle activity throughout the acceleration phase in the curve would help better comprehend the underlying reasons explaining the GRF modifications in the curve.

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DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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